



Koneru Lakshmaiah Education Foundation

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OPERATING SYSTEMS AND SYSTEMS PROGRAMMING (25CS2104E)

2026-27, Term-I

Project Problem Statement Submission Form

Section / Team Details

Section No.: 12

Team No.: 08

Project Title: Linux File System Explorer Using System Calls

Team Members

Roll Number	Student Name	Signature
2520080045	Sindhuja	<i>Sindhuja</i>
2520090121	Jhanasri Thota	<i>Jhanasri</i>
2520090126	Alekhyia	<i>S. Alekhyia</i>

1. Abstract

The **Linux File System Explorer Using System Calls** is a Linux-based command-line application designed to explore and manage files and directories using Linux system calls and POSIX APIs. The project focuses on understanding how a user-space program interacts with the Linux operating system and its file system. The proposed system will allow users to perform basic file-system operations such as listing directory contents, creating files and directories, reading and writing files, viewing file information, and deleting files. Linux APIs and system calls such as `open()`, `read()`, `write()`, `close()`, `mkdir()`, `opendir()`, `readdir()`, `stat()`, and `unlink()` will be used to implement these operations.

The project demonstrates important Operating Systems and System Programming concepts, particularly system calls, file descriptors, file abstraction, directory management, file I/O, and interaction between user space and kernel services. It primarily addresses CO-5: File Systems, File Abstractions, and File I/O in Linux, with a connection to CO-1: The OS as a Service Layer. The expected outcome is a simple and functional terminal-based file explorer that provides practical understanding of Linux file-system operations and system programming.

2. Problem Statement

Linux provides various commands and utilities for managing files and directories, but using these commands alone does not give a clear understanding of how the operating system performs file operations internally. Students need practical exposure to how user programs interact with the Linux kernel through system calls perform basic file and directory operations such as listing directories, creating files and directories, reading and writing files, viewing file information, and deleting files. These operations will be implemented using **Linux system calls and POSIX APIs**.

The project is relevant to Operating Systems and System Programming as it demonstrates concepts such as system calls, file descriptors, file abstraction, directory management, file I/O, and interaction between user space and kernel services, mainly focusing on CO-5 and supporting CO-1.

3. Objectives

1. To develop a Linux-based command-line file system explorer for performing basic file and directory operations.
2. To implement file and directory operations using Linux system calls and POSIX APIs.
3. To understand and demonstrate Operating Systems concepts such as system calls, file descriptors, file abstraction, directory management, and file I/O.
4. To provide practical experience in Linux System Programming and understand how a user-space application interacts with the Linux kernel.

4. Proposed Methodology

The project will be designed as a menu-driven command-line application in C/C++ on Linux/Ubuntu. The application will accept user choices and perform the corresponding file-system operation using Linux system calls and POSIX APIs. The major functionalities will include listing directory contents, creating files and directories, reading and writing files, viewing file information, and deleting files. APIs such as `open()`, `read()`, `write()`, `close()`, `mkdir()`, `opendir()`, `readdir()`, `stat()`, and `unlink()` will be used.

The basic flow of the system will be:

User → Command-Line Interface → System Call/POSIX API → Linux Kernel → File System → Result displayed to User

The program will also check system-call return values and display appropriate error messages for invalid files, directories, paths, or permissions.

5. Operating Systems Concepts / Linux APIs Used

OS Concept / Linux API / System Call	Purpose in the Project
System Calls	Provides the interface between the user-space application and the Linux kernel.
File Descriptors	Used to identify and access opened files during file I/O operations.
<code>open()</code>	Opens or creates a file and returns a file descriptor.
<code>read()</code>	Reads data from an opened file.
<code>write()</code>	Writes data into a file.
<code>close()</code>	Closes an opened file and releases its resources.
<code>mkdir()</code>	Creates a new directory.
<code>opendir() / readdir()</code>	Opens a directory and reads its entries for listing files and folders.
<code>stat()</code>	Retrieves file information such as size, type, and permissions.
<code>unlink()</code>	Deletes a file from the file system.

6. Individual Contribution

Roll Number	Student Name	Individual Responsibility
2520080045	Sindhuja	Project design, menu-driven interface, file creation, reading, writing, and overall integration
2520090121	Jhanasri Thota	Directory operations using <code>mkdir()</code> , <code>opendir()</code> , <code>readdir()</code> , and <code>closedir()</code>
2520090126	Alekhya	File information and deletion using <code>stat()</code> and <code>unlink()</code> , error handling, testing, documentation, and GitHub management

7. Tools / Platforms / Software Used

Tool / Platform / Software	Purpose
Linux / Ubuntu	Operating system environment for developing and running the project.
C / C++	Programming language used to implement the file system explorer.
GCC Compiler	Compiles and builds the C/C++ source code.
Linux System Calls / POSIX APIs	Used to perform file and directory operations.
Git / GitHub	Used for version control and storing the project source code.

8. Expected Outcome

The project is expected to produce a functional command-line Linux File System Explorer that allows users to perform basic file and directory operations using Linux system calls and POSIX APIs. The application will allow users to list directory contents, create files and directories, read and write file contents, view file information such as size, type and permissions, and delete files. It will also provide appropriate error messages for invalid operations. The project will demonstrate practical understanding of system calls, file descriptors, file abstraction, directory management, and file I/O, mainly supporting CO-5 and CO-1 of the OSSP syllabus.

9. GitHub Repository

GitHub Repository Link: Jhanasri-thota/KLH_CSIT-AIDS_2026-27_Team08_LinuxFileSystemExplorer

Repository Created: YES

- ☐ Yes
- ☐ No

Faculty Name and Signature

Faculty Name: Raghupathi Manthena

Signature: _____

Remarks: _____

Faculty Use

Project Approved: _____ yes

☒ Yes ☐ No

Date: _____

Faculty Signature: _____